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A Pilot Study on the Effect of Educational Intervention on Nurses' Knowledge Towards Insulin Therapy for Diabetes Mellitus

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Introduction

The International Diabetes Federation Diabetes Atlas 2025 reported that **21.1%** of the Malaysians were living with diabetes in 2024.

Glycaemic control with insulin therapy is essential to reduce diabetes-related complications and **correct insulin injection technique** is critical for clinical effectiveness.

Previous studies identified gaps in nurses' knowledge of insulin injection technique, potentially contributing to administration errors and adverse outcomes.

Objective

To design and evaluate the **effect of educational intervention** among registered nurses of Batu Gajah Hospital in **insulin-related knowledge and insulin pen injection technique** in order to improve patient care and safety.

Materials and Methods

Study Design
Single-cohort pre-post pilot study in August 2023

Setting
Hospital Batu Gajah (HBG), Perak

Participants
Convenience sampling of registered nurses HBG (excluding Matron U36 and registered nurses on vacation/further studies)
N = 46

Intervention
Two **Educational Workshop sessions** (PowerPoint presentation + Discussion) with **hands-on training** adapted from the Forum for Injection Technique, Malaysia Region (FIT-MY) Training-of-Trainers (ToT) Programme

Data Collection
To complete a 13-item **Insulin-related Knowledge Questionnaire** adapted from the FIT-MY ToT Questionnaire + **Insulin injection technique assessment** as the baseline measurement before and immediately after the educational intervention workshop

Analysis
Wilcoxon signed-rank test for pre-post intervention knowledge score.
Paired t-test for pre-post insulin injection technique assessment score.

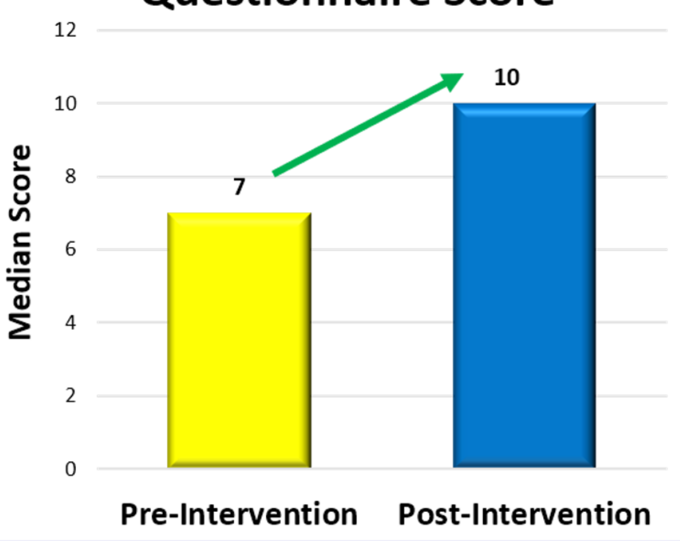
Results

Demographic	n (%)
Gender	
Female	45 (97.8)
Male	1 (2.2)
Age (years), Mean (SD)	42.48 (6.32)
Years in Practice	
≤ 3	5 (10.9)
4 – 7	5 (10.9)
8 - 11	10 (21.7)
≥ 11	26 (56.5)
Current Department	
Medical	15 (32.6)
O&G	6 (13.0)
Paediatrics	5 (10.9)
Medical	15 (32.6)
Others	20 (43.5)

Pre-Intervention Score		
Baseline Analysis	**Result**	
Kruskal-Wallis test: Departments vs Score		
Knowledge Questionnaire	p = 0.687	No Significant Difference
Injection Technique	p = 0.537	
Spearman's rho: Years of Clinical Experience vs Score		
Knowledge Questionnaire	p = 0.769	No Significant Correlation
Injection Technique	p = 0.411	

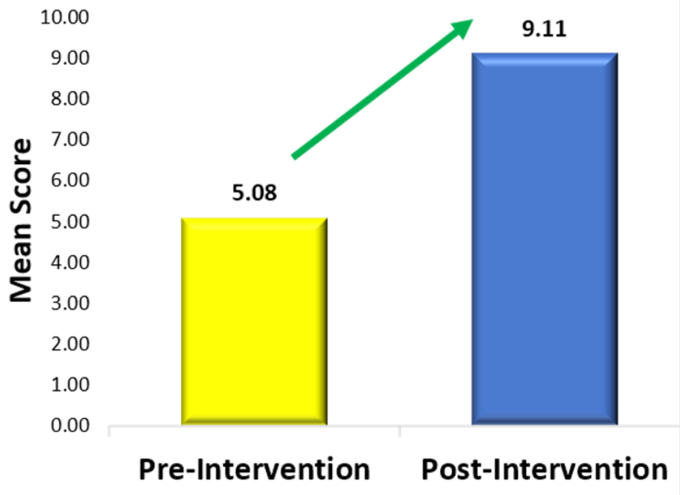
Participants Recruited, N = 46
Post intervention:
Insulin-related Questionnaire, n = 45
Injection Technique Assessment, n = 38

Insulin-Related Knowledge Questionnaire Score



Significant improvement in knowledge score after educational intervention. (Wilcoxon signed-rank test: Z=−4.196, *p*<0.001)

Insulin Injection Technique Assessment Score



Significant improvement in injection technique score after educational intervention. (Paired t-test, *p*<0.001)

Summary of Key Findings

- Significant improvement in questionnaire and injection technique scores post-intervention.
- Greatest improvements in knowledge were observed regarding symptoms of hypoglycaemia and reasons for patients' reluctance to use insulin.
- No significant differences in baseline pre-intervention questionnaire and injection technique score across different departments.
- No significant correlations between years of clinical experience and baseline scores for both questionnaire and injection technique.

Discussion

The median pre-intervention questionnaire score was 7 out of 13, indicating a critical knowledge gap, with general participating nurses answering only approximately half of the items correctly.

The significant post-intervention improvements of the study were consistent with previous studies, demonstrating that structured educational intervention with hands-on training effectively improved insulin-related knowledge and injection technique.

The mean technique assessment score was 5.08 out of 11, indicating suboptimal adherence of nurses' insulin injection practices to recommended insulin injection technique guidelines.

No significant correlations were observed between years of clinical experience or department category and either baseline questionnaire scores or technique assessment scores, highlighting the need for standardized insulin education for all nurses regardless of seniority or clinical area.

Conclusion

Structured educational intervention workshop with hands-on training **significantly improved** nurses' insulin-related knowledge and injection technique, supporting its potential to enhance nurses' competency.

Clinical Implications: Improved nurses' competency in insulin pen injection may facilitate supervised patient self-administration of insulin during hospitalization, reinforce correct injection technique and potentially improve patients' treatment adherence and clinical outcomes.

Study limitations: Convenience sampling, participant attrition due to time or work constraints, and the absence of follow-up assessment may limit the evaluation of the sustainability of the intervention effect.
Future studies should assess long-term retention of knowledge and technique as well as intervention's impact on patient clinical outcomes.

Acknowledgement

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Conflict of Interest:
The authors declare no conflict of interest related to this study.

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